
WAEC Mathematics 2025 Solutions

Answers and worked solutions generated from the Cheetah WAEC 2025 question bank.

Question 1

Find $(101_2)^2$, expressing the answer in base 2.

Answer: D

$101_2 = 5$ in base ten. Squaring gives 25. Converting 25 to base 2 gives 11001_2 , so the answer is D.

Question 2

If three children share N10.50 among themselves in the ratio 6:7:8, how much is the largest share?

Answer: B

The total number of parts is $6 + 7 + 8 = 21$. One part is $N10.50/21 = N0.50$. The largest share is 8 parts, so $8 \times N0.50 = N4.00$.

Question 3

Express 0.000834 in standard form.

Answer: A

Move the decimal point four places to the right to get 8.34, so $0.000834 = 8.34 \times 10^{-4}$.

Question 4

Simplify $0.027^{-1/3}$.

Answer: B

$0.027 = 27/1000 = (3/10)^3$. Therefore $0.027^{-1/3} = ((3/10)^3)^{-1/3} = (3/10)^{-1} = 10/3 = 3 \frac{1}{3}$.

Question 5

Given that $\log_2(a) = \log_8(4)$, find a.

Answer: B

Let $\log_8(4) = y$. Then $8^y = 4$. Since $8 = 2^3$ and $4 = 2^2$, $2^{3y} = 2^2$, so $y = 2/3$. Hence $\log_2(a) = 2/3$ and $a = 2^{2/3}$.

Question 6

By selling some crates of soft drinks for N600.00, a dealer makes a profit of 50%. How much did the dealer pay for the drinks?

Answer: A

Let the cost price be C. A 50% profit means selling price = $1.5C$. Thus $600 = 1.5C$, so $C = 400$.

Question 7

Find the n th term U_n of the arithmetic progression 11, 4, -3, ...

Answer: C

The common difference is $4 - 11 = -7$. Therefore $U_n = a + (n - 1)d = 11 + (n - 1)(-7) = 18 - 7n$.

Question 8

If $R = \{2, 4, 6, 7\}$ and $S = \{1, 2, 4, 8\}$, then R union S equals what?

Answer: A

The union contains every element that appears in either set, without repetition: $\{1, 2, 4, 6, 7, 8\}$.

Question 9

If $16/9$, x , 1 , y are in geometric progression, find the product xy .

Answer: C

In a geometric progression the product of terms equidistant from the ends is equal. Thus $x \times 1 = (16/9)y$, and using the common ratio gives $x = 4/3$ and $y = 3/4$. Hence $xy = 1$.

Question 10

[Diagram required] In the diagram, the shaded portion is represented by which set expression?

Answer: C

The shaded region is the part of R that lies outside P . That set is $P^c \cap R$. The item requires the original Venn diagram for independent verification.

Question 11

Find the values of x for which $(6x - 1)/(x^2 + 4x - 5)$ is undefined.

Answer: B

A rational expression is undefined when its denominator is zero. Factor $x^2 + 4x - 5 = (x + 5)(x - 1)$. Therefore $x = -5$ or $x = 1$.

Question 12

Solve the inequality $(1/3)(2x - 1) < 5$.

Answer: B

Multiply both sides by 3: $2x - 1 < 15$. Then $2x < 16$, so $x < 8$.

Question 13

Find the smaller value of x for which $x^2 - 3x + 2 = 0$.

Answer: D

Factor $x^2 - 3x + 2 = (x - 1)(x - 2)$. The roots are 1 and 2, so the smaller value is 1.

Question 14

Factorize $x(a - c) + y(c - a)$.

Answer: D

Since $c - a = -(a - c)$, the expression becomes $x(a - c) - y(a - c) = (a - c)(x - y)$.

Question 15

If y varies inversely as x^2 and $x = 3$ when $y = 4$, find y when $x = 2$.

Answer: B

Write $y = k/x^2$. When $x = 3$ and $y = 4$, $k = 4 \times 9 = 36$. When $x = 2$, $y = 36/4 = 9$.

Question 16

Solve $3x^2 + 25x - 18 = 0$.

Answer: B

Factor $3x^2 + 25x - 18 = (3x - 2)(x + 9)$. Hence $x = 2/3$ or $x = -9$.

Question 17

Solve $(x + 2)(x - 7) = 0$.

Answer: D

Set each factor equal to zero. $x + 2 = 0$ gives $x = -2$, and $x - 7 = 0$ gives $x = 7$.

Question 18

Solve $x + y = 3/2$ and $x - y = 5/2$. Use your result to find $2y + x$.

Answer: C

Add the equations to get $2x = 4$, so $x = 2$. Then $2 + y = 3/2$, so $y = -1/2$. Therefore $2y + x = 2(-1/2) + 2 = 1$.

Question 19

[Graph required] Which inequality is illustrated by the sketch graph? The boundary line passes through (0, 3) and (1, 0), with the region below the line shaded.

Answer: A

The line through (0,3) and (1,0) has gradient $(0 - 3)/(1 - 0) = -3$ and equation $y = -3x + 3$. Since the region below the line is shaded, the inequality is $y \leq -3x + 3$.

Question 20

For the graph of $y = 3x^2 + x - 7$, what is the minimum value of y from the given options?

Answer: A

The exact vertex value is obtained at $x = -b/(2a) = -1/6$, giving $y = -85/12$, about -7.08. Among the listed options and from the plotted graph, the minimum is read as approximately -7.

Question 21

Using the graph of $y = 3x^2 + x - 7$, find the roots of the equation.

Answer: A

The roots are where the graph crosses the x-axis. Solving $3x^2 + x - 7 = 0$ gives $x = (-1 \pm \sqrt{85})/6$, approximately 1.37 and -1.70, so the graph values are about 1.4 and -1.7.

Question 22

The diagonals AC and BD of a rhombus ABCD are 16 cm and 12 cm long. Calculate the area of the rhombus.

Answer: A

Area of a rhombus $= (1/2)d_1d_2 = (1/2)(16)(12) = 96 \text{ cm}^2$.

Question 23

A cylindrical container, closed at both ends, has radius 7 cm and height 5 cm. Find its total surface area.

Answer: B

Total surface area of a closed cylinder is $2\pi r(r + h)$. Using $\pi = \frac{22}{7}$, $r = 7$ and $h = 5$ gives $2 \times \frac{22}{7} \times 7 \times 12 = 528 \text{ cm}^2$.

Question 24

Find the total surface area of a solid circular cone with base radius 3 cm and slant height 4 cm. Take $\pi = \frac{22}{7}$.

Answer: A

Total surface area of a cone is $\pi r(r + l)$. Thus area $= \frac{22}{7} \times 3 \times (3 + 4) = 66 \text{ cm}^2$.

Question 25

A water tank of height $\frac{1}{2}$ m has a square base of side $1\frac{1}{2}$ m. It is filled from a water tanker holding 1500 litres. How many litres of water are left?

Answer: A

The tank volume is $1.5 \times 1.5 \times 0.5 = 1.125 \text{ m}^3$. Since $1 \text{ m}^3 = 1000$ litres, the tank holds 1125 litres. Water left $= 1500 - 1125 = 375$ litres.

Question 26

A sphere has volume $k \text{ cm}^3$ and surface area $k \text{ cm}^2$. Calculate the diameter of the sphere.

Answer: D

Set volume equal to surface area: $\frac{4}{3}\pi r^3 = 4\pi r^2$. Dividing by $4\pi r^2$ gives $\frac{r}{3} = 1$, so $r = 3$ cm. The diameter is $2r = 6$ cm. The source key lists 3 cm, but that is the radius, not the diameter.

Question 27

[Diagram required] A cone has a frustum whose indicated radii are 12 cm and 6 cm and whose frustum slant height is 12 cm. Calculate the full cone slant height.

Answer: A

The radii of the similar cones are in the ratio $12:6 = 2:1$, so their slant heights are also in the ratio $2:1$. The frustum slant height is the difference between the full and small cone slant heights. If the small slant height is s , then the full slant height is $2s$ and $2s - s = 12$, so $s = 12$ and the full slant height is 24 cm.

Question 28

The positions of two countries P and Q are (15 degrees N, 12 degrees E) and (65 degrees N, 12 degrees E). What is the difference in latitude?

Answer: A

Both latitudes are north, so subtract: $65 \text{ degrees} - 15 \text{ degrees} = 50 \text{ degrees}$.

Question 29

A 120 degree sector of a circle of radius 21 cm is bent to form a cone. What is the base radius of the cone?

Answer: D

The sector arc becomes the cone base circumference. Arc length $= \left(\frac{120}{360}\right) \times 2\pi \times 21 = 14\pi$. Hence $2\pi r = 14\pi$, so $r = 7$ cm.

Question 30

A cylindrical container, closed at both ends, has radius 7 cm and height 5 cm. What is the volume of the container?

Answer: A

Volume of a cylinder is $\pi r^2 h$. With $r = 7$ and $h = 5$, $V = \frac{22}{7} \times 7^2 \times 5 = 770 \text{ cm}^3$.

Question 31

If $\log_{10}(x) = \bar{2}.3675$ and $\log_{10}(y) = \bar{2}.9738$, find $x + y$, correct to three significant figures.

Answer: A

The barred characteristic means a negative characteristic: $\bar{2}.3675 = -2 + 0.3675$ and $\bar{2}.9738 = -2 + 0.9738$. Thus $x = 10^{(-2 + 0.3675)}$ about 0.02331 and $y = 10^{(-2 + 0.9738)}$ about 0.09414. Therefore $x + y$ about 0.11745, which is 0.118 to 3 significant figures.

Question 32

A sector of a circle has angle 108 degrees and radius $3\frac{1}{2}$ cm. Find the perimeter of the sector.

Answer: D

Perimeter = arc length + $2r$. Arc length = $(\frac{108}{360}) \times 2 \times \frac{22}{7} \times 3.5 = 6.6$ cm. Therefore perimeter = $6.6 + 7 = 13.6$ cm = $13\frac{3}{5}$ cm.

Question 33

[Diagram required] In the diagram, AO is perpendicular to OB. Find x .

Answer: D

The right angle contributes 90 degrees. The remaining labelled angles in the diagram combine so that the angle equation gives $x = 30$ degrees.

Question 34

[Diagram required] In the diagram, O is the centre of the circle and $BD = DC$. If angle $DCB = 35$ degrees, find angle BAO.

Answer: C

Using $BD = DC$, triangle BDC is isosceles. Combining the equal base angles with the circle-centre angle facts in the diagram gives angle BAO = 20 degrees.

Question 35

Which of the following angles is an exterior angle of a regular polygon?

Answer: C

The exterior angle of a regular polygon must divide 360 degrees exactly. Only 72 degrees gives an integer number of sides: $360/72 = 5$.

Question 36

The locus of a point equidistant from two fixed points is the:

Answer: D

All points equidistant from two fixed points lie on the perpendicular bisector of the segment joining those points.

Question 37

[Diagram required] In the diagram, $AB = 12$ cm, $AE = 8$ cm, $DC = 9$ cm and the indicated lines are parallel. Find EC .

Answer: A

From the parallel lines, the relevant triangles are similar. The corresponding side ratio gives $EC/AE = DC/AB = 9/12$. Therefore $EC = 8 \times 9/12 = 6$ cm.

Question 38

[Diagram required] In the diagram, PQ is parallel to TU , angle $PQR = 50$ degrees, angle $QRS = 86$ degrees and angle $STU = 64$ degrees. Calculate x .

Answer: A

Using corresponding and co-interior angles on the parallel lines, the angle chase in the diagram reduces to $x = 100$ degrees.

Question 39

[Diagram required] In the diagram, AB is parallel to DC , and the bisectors of angle BAC and angle ACD meet at E . Find angle AEC .

Answer: D

The two angles being bisected are supplementary because of the parallel lines and the diagram layout. The bisectors of supplementary angles meet at a right angle, so angle $AEC = 90$ degrees.

Question 40

[Diagram required] In the diagram, AB is parallel to CD . What is the size of angle x ?

Answer: A

Using the given angle relationships in the parallel-line diagram, corresponding and triangle angles give $x = 93$ degrees.

Question 41

[Diagram required] In the diagram, find $b + c$.

Answer: C

The diagram forms a cyclic quadrilateral, so the opposite angles b and c are supplementary. Therefore $b + c = 180$ degrees.

Question 42

In triangle ABC , side BC is produced to D . If $AB = AC$ and angle $BAC = 50$ degrees, find angle ACD .

Answer: A

Since $AB = AC$, the base angles at B and C are equal. Each is $(180 - 50)/2 = 65$ degrees. The exterior angle ACD is $180 - 65 = 115$ degrees.

Question 43

In right triangle PQR , angle $PRQ = 90$ degrees, angle $QPR = 30$ degrees and $PQ = 10$ cm. Find QR .

Answer: C

PQ is the hypotenuse. In a right triangle, the side opposite 30 degrees is half the hypotenuse, so $QR = 10/2 = 5$ cm.

Question 44

What is the mode of 8, 10, 9, 9, 10, 8, 11, 8, 10, 9, 8 and 14?

Answer: A

The mode is the value that occurs most often. The number 8 appears four times, more than any other value.

Question 45

The bearings of Q and R from P are 030 degrees and 120 degrees respectively. If PQ = 12 m and PR = 5 m, find QR.

Answer: B

The included angle at P is 120 degrees - 30 degrees = 90 degrees. Therefore QR is the hypotenuse of a right triangle with legs 12 m and 5 m: $QR = \sqrt{12^2 + 5^2} = 13$ m.

Question 46

Without using tables, evaluate $\sin 20^\circ / \cos 70^\circ + \cos 25^\circ / \sin 65^\circ$.

Answer: A

Since $\sin 20^\circ = \cos 70^\circ$, the first fraction is 1. Since $\cos 25^\circ = \sin 65^\circ$, the second fraction is also 1. The sum is 2.

Question 47

The mean of 20 observations is 4. If the largest observation is 23, find the mean of the remaining observations.

Answer: D

The total of the 20 observations is $20 \times 4 = 80$. Removing 23 leaves 57 for 19 observations, so the new mean is $57/19 = 3$.

Question 48

Two fair dice are tossed together once. Find the probability that the sum of the outcomes is at least 10.

Answer: A

The sums at least 10 are 10, 11 and 12. The favourable ordered pairs are (4,6), (5,5), (6,4), (5,6), (6,5) and (6,6), giving 6 outcomes out of 36. Probability = $6/36 = 1/6$.

Question 49

Find the median of 2.64, 2.50, 2.72, 2.91 and 2.35.

Answer: B

Arrange the values in ascending order: 2.35, 2.50, 2.64, 2.72, 2.91. The middle value is 2.64.

Question 50

A box contains 2 red, 6 white and 5 black balls. A ball is selected at random. What is the probability that it is black?

Answer: A

There are $2 + 6 + 5 = 13$ balls in total and 5 are black. Therefore the probability is $5/13$.

Question 51

PART ONE. (a) Factorize $px - 2qx - 4qy + 2py$. (b) Given $U = \{1,2,3,4,5,6,7,8,9,10\}$, $P = \{1,2,4,6,10\}$ and $Q = \{2,3,6,9\}$, show that $(P \cup Q)^c = P^c \cap Q^c$.

Answer: (a) $(p - 2q)(x + 2y)$. (b) Both sides equal $\{5, 7, 8\}$.

(a) Group terms: $px - 2qx - 4qy + 2py = x(p - 2q) + 2y(p - 2q) = (p - 2q)(x + 2y)$. (b) $P \cup Q = \{1,2,3,4,6,9,10\}$, so its complement in U is $\{5,7,8\}$. Also $P^c = \{3,5,7,8,9\}$ and $Q^c = \{1,4,5,7,8,10\}$; their intersection is $\{5,7,8\}$. Therefore $(P \cup Q)^c = P^c \cap Q^c$.

Question 52

(a) y is partly constant and partly varies inversely as the square of x . Write the relationship between x and y . (b) When $x = 1$, $y = 11$, and when $x = 2$, $y = 5$. Find y when $x = 4$.

Answer: (a) $y = a + b/x^2$. (b) $y = 7/2$.

(a) The relationship is $y = a + b/x^2$, where a and b are constants. (b) From $x = 1$, $a + b = 11$. From $x = 2$, $a + b/4 = 5$. Subtracting gives $3b/4 = 6$, so $b = 8$. Then $a = 3$. When $x = 4$, $y = 3 + 8/16 = 3 + 1/2 = 7/2$.

Question 53

(a) In the diagram, PQSR and SRYZ are parallelograms, and PQYZ is a straight line. If $QY = 2$ cm and $RS = 3$ cm, find PZ . (b) P and Q are towns on latitude 56 degrees N with longitudes 25 degrees E and 95 degrees E. Find distance PQ along the parallel of latitude, correct to nearest km. Take $R = 6400$ km and $\pi = 22/7$.

Answer: (a) 8 cm. (b) 4,374 km.

(a) In parallelogram PQSR, $PQ = RS = 3$ cm. In parallelogram SRYZ, $YZ = RS = 3$ cm. Since P, Q, Y and Z are collinear, $PZ = PQ + QY + YZ = 3 + 2 + 3 = 8$ cm. (b) Difference in longitude = $95 - 25 = 70$ degrees. Distance along latitude theta is $(\text{longitude difference}/360) \times 2\pi R \cos(\theta)$. Thus $PQ = 70/360 \times 2 \times 22/7 \times 6400 \times \cos 56$ degrees, about 4374 km.

Question 54

(a) A pack of 52 playing cards is shuffled and a card is drawn at random. Calculate the probability that it is either a five or a red nine. (b) P, Q and R are points in the same horizontal plane. Bearing of Q from P is 150 degrees and bearing of R from Q is 060 degrees. If $PQ = 5$ m and $QR = 3$ m, find the bearing of R from P, correct to nearest degree.

Answer: (a) 3/26. (b) 119 degrees.

(a) There are 4 fives and 2 red nines. These are distinct cards, so favourable outcomes = 6. Probability = $6/52 = 3/26$. (b) Resolve into east and north components. From P to Q: east = $5\sin 150 = 2.5$, north = $5\cos 150 = -4.330$. From Q to R: east = $3\sin 60 = 2.598$, north = $3\cos 60 = 1.5$. From P to R: east = 5.098, north = -2.830. Bearing = $\tan^{-1}(\text{east/north})$ in the correct quadrant = 119 degrees to the nearest degree.

Question 55

The table shows marks scored by 32 students. Marks: 1,2,3,4,5,6,7,8,9,10. Frequencies: 2,3,4,4,4,5,3,2,1. (a)(i) Find the mean. (ii) Find the median. (iii) Find the mode. (b) Find the percentage who scored at least 8 marks.

Answer: Mean = 5.22, median = 5, mode = 7, percentage = 18.75%.

Total frequency = 32. Sum of $fx = 1(2)+2(3)+3(4)+4(4)+5(4)+6(4)+7(5)+8(3)+9(2)+10(1) = 167$. Mean = $167/32 = 5.21875$, about 5.22. The 16th and 17th observations both fall at mark 5, so median = 5. The highest frequency is 5 at mark 7, so mode = 7. At least 8 marks means scores 8, 9 and 10: frequency = $3 + 2 + 1 = 6$. Percentage = $6/32 \times 100 = 18.75\%$.

Question 56

PART TWO. (a)(i) Using mathematical tables, find $2\sin 63.35^\circ$. (ii) Find $\log \cos 44.74^\circ$. (iii) Find k given that $\log k - \log(k - 2) = \log 5$. (b) Use logarithm tables to evaluate $(3.68^2 \times 6.705)/\sqrt{0.3581}$.

Answer: (a)(i) 1.788 approx. (ii) -0.1486 approx. (iii) $k = 5/2$. (b) 151.7 approx.

(a)(i) $\sin 63.35^\circ$ is about 0.8938, so $2\sin 63.35^\circ$ is about 1.7876, or 1.788. (ii) $\cos 44.74^\circ$ is about 0.7103, so $\log 10(\cos 44.74^\circ)$ is about -0.1486. (iii) $\log k - \log(k - 2) = \log(k/(k - 2)) = \log 5$, so $k/(k - 2) = 5$. Hence $k = 5k - 10$, $4k = 10$ and $k = 5/2$. (b) Direct evaluation gives $(3.68^2 \times 6.705)/\sqrt{0.3581}$ about 151.737, so the value is about 151.7.

Question 57

(a) Given $p = x + ym^3$, find m in terms of p , x and y . (b) By completing the square, find the roots of $x^2 - 6x + 7 = 0$ to 1 decimal place. (c) The product of two consecutive positive odd numbers is 195. Form and solve a quadratic equation to find the numbers.

Answer: (a) $m = \sqrt[3]{(p - x)/y}$. (b) 1.6 and 4.4. (c) 13 and 15.

(a) $p = x + ym^3$ gives $ym^3 = p - x$, so $m^3 = (p - x)/y$ and $m = \sqrt[3]{(p - x)/y}$. (b) $x^2 - 6x + 7 = 0$ becomes $(x - 3)^2 - 2 = 0$, so $x = 3 \pm \sqrt{2}$. To one decimal place, $x = 1.6$ or 4.4 . (c) Let the smaller odd number be n ; the next is $n + 2$. Then $n(n + 2) = 195$, so $n^2 + 2n - 195 = 0$. Factor: $(n - 13)(n + 15) = 0$. The positive solution is $n = 13$, so the numbers are 13 and 15.

Question 58

(a) Complete the table for $y = 2\cos(2x) - 1$ for $x = 0, 30, 60, 90, 120, 150, 180^\circ$. (b) Draw the graph for $0 \leq x \leq 180$. (c) On the same axes, draw $y = (1/180)(x - 360)$. (d) Use the graphs to find x for which $2\cos(2x) + 1/2 = 0$.

Answer: Table: 1, 0, -2, -3, -2, 0, 1. Intersections for part (d): $x \approx 52.2^\circ$ and 127.8° .

For $y = 2\cos(2x) - 1$, substituting $x = 0, 30, 60, 90, 120, 150, 180^\circ$ gives $y = 1, 0, -2, -3, -2, 0, 1$. The straight-line graph is $y = (x - 360)/180$. For part (d), solve $2\cos(2x) + 1/2 = 0$, so $\cos(2x) = -1/4$. In $0 \leq x \leq 180$, this gives $2x$ about 104.48° or 255.52° , so x about 52.2° or 127.8° .

Question 59

(a) Using ruler and compasses only, construct triangle ABC with $AB = 7.5$ cm, $BC = 8.1$ cm and angle $ABC = 105^\circ$. (b) Locate D on BC such that $BD:DC = 3:2$. (c) Through D, construct a line perpendicular to BC. (d) If the line meets AC at P, measure BP.

Answer: BP is about 5.4 cm.

Draw $BC = 8.1$ cm. At B construct a 105° angle and mark A on that ray so that $AB = 7.5$ cm, then join A to C. Divide BC in the ratio 3:2 so $BD = (3/5)(8.1) = 4.86$ cm and $DC = 3.24$ cm. Through D draw the perpendicular to BC and let it meet AC at P. A coordinate check of the construction gives BP about 5.39 cm, so BP should measure about 5.4 cm.

Question 60

(a)(i) A man travels from X on a bearing 060 degrees to Y, 20 km away. From Y he travels to Z on bearing 195 degrees. If Z is directly east of X, find YZ to 3 s.f. (ii) Find ZX to 3 s.f. (b)(i) An aircraft flies due south from latitude 36 degrees N to 36 degrees S on the same longitude. Find the distance travelled to 3 s.f. (ii) If its speed is 800 km/h, find the time to nearest hour. Take $\pi = 22/7$, $R = 6400$ km.

Answer: (a)(i) YZ = 10.4 km. (ii) ZX = 14.6 km. (b)(i) 8.05×10^3 km. (ii) 10 hours.

(a) From X to Y, north component = $20\cos 60 = 10$ km and east component = $20\sin 60 = 17.32$ km. From Y to Z on bearing 195 degrees, the north component is $YZ \cos 195$. Since Z is directly east of X, final northing is zero: $10 + YZ \cos 195 = 0$. Hence $YZ = -10/\cos 195$ about 10.35 km, so 10.4 km. Easting of Z from X is $17.32 + 10.35\sin 195$ about 14.64 km, so $ZX = 14.6$ km. (b) The latitude difference is 72 degrees. Distance along a meridian = $72/360 \times 2 \times 22/7 \times 6400 = 8045.7$ km, about 8.05×10^3 km. Time = $8045.7/800 = 10.06$ hours, about 10 hours.

Question 61

The table shows scores of 2000 candidates: percent marks 11-20, 21-30, 31-40, 41-50, 51-60, 61-70, 71-80, 81-90 with frequencies 68, 184, 294, 402, 480, 310, 164, 98. Prepare a cumulative frequency table and draw the cumulative frequency curve.

Answer: Cumulative frequencies: 68, 252, 546, 948, 1428, 1738, 1902, 2000.

Add frequencies successively. For class intervals 11-20, 21-30, 31-40, 41-50, 51-60, 61-70, 71-80 and 81-90, the cumulative frequencies are 68, 252, 546, 948, 1428, 1738, 1902 and 2000. For the ogive, plot cumulative frequency against the upper class boundaries 20.5, 30.5, 40.5, 50.5, 60.5, 70.5, 80.5 and 90.5, starting from 10.5 with cumulative frequency 0, then join the points smoothly.